Fine-Tuning a 7B Math LLM for Olympiad Reasoning

A Diagnostic Study of Evaluation Methodology and OOD Generalization

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Abstract

We fine-tune `Qwen2.5-Math-7B-Instruct` with LoRA on 72,283 filtered olympiad problems and evaluate on val_200 (n=200) and AIME 2024+2025 (n=60). Three evaluation bugs produced a **12-point swing** in the measured SFT effect, turning an apparent -2-point regression into a genuine $+10$-point gain. Under corrected evaluation, Round 1 (r=64, lr= 2e$-$4) achieves 67% on val_200 but degrades on AIME 2024 (-6.7 pts), a catastrophic-forgetting signature. A hyperparameter-only Round 2 (r=16, lr= 1e$-$5) produces an asymmetric OOD result that implicates a persistent training/inference format mismatch as the dominant cause.

1 Setup

Model and data. We fine-tune `Qwen/Qwen2.5-Math-7B-Instruct` [4] (7B, already RLHF-aligned) with LoRA [2] adapters on all seven projection matrices. Training data is NuminaMath-CoT + TIR [3] filtered to olympiad-level sources with integer answers (72,283 train examples, 95/5 split, seed 42).

Training format mismatch (key design flaw). Both rounds train using a plain-text template (`"Problem: {p}\n\nSolution: {s}"`), while evaluation uses Qwen's ChatML template. This mismatch is identified post-hoc as the likely dominant cause of OOD degradation.

Hyperparameters. Table 1 summarizes the two rounds; all else is identical (1 epoch, effective batch 32, cosine schedule).

Table 1: Training hyperparameters by round.

	Round 1	Round 2
LoRA rank r	64	16
LoRA α	128	32
Learning rate	2e-4	1e-5
Precision	fp16	bf16

Evaluation. Greedy generation, last `\boxed{...}` extracted by regex, numeric-tolerant equality ("5.0" $\equiv$ "5"), `max_new_tokens=2048`, ChatML prompt format. All reported numbers use this corrected protocol (eval v2).

2 Results

2.1 Main Results

Table 2: Accuracy under corrected evaluation. Wilson 95% CIs in brackets.

Benchmark	Baseline	Rnd 1	Rnd 2
val_200	57.0 [53.6, 60.4]	**67.0** [63.6, 70.4]	55.0 [48.0, 61.7]
AIME 2024	23.3 [11.8, 40.9]	16.7 [7.3, 33.6]	3.3 [0.6, 16.7]
AIME 2025	6.7 [1.6, 21.6]	6.7 [1.6, 21.6]	10.0 [3.5, 25.6]

Round 1 gains +10 points on val_200 ($p < 0.01$) but loses 6.7 points on AIME 2024. Round 2's smaller adapter reduces in-distribution specialization (val_200 drops to 55%) while producing the asymmetric OOD pattern described in Section 3.

2.2 Evaluation Methodology Finding

Table 3: The 12-point methodological swing on val_200.

Model	Eval v1 (buggy)	Eval v2 (fixed)
Baseline	36.0%	57.0%
SFT Round 1	34.0%	67.0%
Δ	-2.0 pts	$+10.0$ pts

Three bugs in eval v1 collectively account for the swing: (1) `max_new_tokens=512` truncated solutions before `\boxed{}` (olympiad solutions run 600–900 tokens); (2) strict string equality rejected numerically equivalent answers like `"5.0"` vs. `"5"`; (3) plain-text prompt format at inference mismatched the ChatML distribution under which Qwen's `\boxed{}` convention was trained. Inspecting 10 raw outputs surfaced all three bugs in $<$15 minutes of free compute.

2.3 Failure-Mode Breakdown

Across all benchmarks, the dominant failure category is `wrong_numeric` (model produces a boxed integer but it is incorrect): 82% of failures on AIME for Round 1, 90% on AIME 2024 for Round 2. Only 1–13% of failures are format failures

(no_box, truncated_no_box), indicating the bottleneck is reasoning capability rather than output formatting. Round 2 shows elevated wrong_nonnumeric on val_200 (12%), indicating partial regression in \boxed{} formatting discipline relative to Round 1.

3 Analysis

Why does Round 1 degrade on AIME 2024? Three mechanisms are consistent with the data: (M1) aggressive LR (2e-4) overwrites RLHF alignment; (M2) full-sequence loss dilutes gradient signal on solutions; (M3) training/inference format mismatch. Round 2 tests M1 in isolation by conserving the hyperparameters while leaving M2 and M3 unfixed.

Round 2 asymmetry rules out M1 as the primary cause. If hyperparameter aggressiveness were dominant, Round 2 should have reduced AIME 2024 forgetting. Instead AIME 2024 collapsed further (16.7% $\rightarrow$ 3.3%). Meanwhile AIME 2025 directionally improved (6.7% $\rightarrow$ 10.0%), consistent with a less-specialized adapter that overfits the wrong format less—benefiting where the base had no prior competence, while further disrupting where it did (AIME 2024, val_200).

This asymmetry implicates M3 as the dominant cause. A larger adapter (Round 1, r=64) can absorb the format mismatch by force-fitting. A smaller adapter (Round 2, r=16) cannot, and perturbs the base model's alignment without compensating in-distribution gain. Conservative hyperparameters are appropriate only *given* a correctly formatted training pipeline.

4 Conclusion and Future Work

The central lesson is methodological: three evaluation bugs produced a 12-point swing in the measured SFT effect, enough to reverse the apparent sign of the intervention. Diagnostic inspection of raw outputs is the cheapest, highest-value step before any retraining decision.

On the modeling side, the structural fix for a Round 3 is straightforward: (1) replace the plain-text training format with tokenizer.apply_chat_template; (2) apply completion-only loss masking via DataCollatorForCompletionOnlyLM. Round 2's hyperparameters (r=16, lr= 1e$-$5, bf16) are retained—they are appropriate given a correct format. Beyond retraining, inference-time tool integration (Python REPL via NuminaMath-TIR) is the most promising direction by literature precedent [1].

References

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